



PROGRAMMED MINI WEATHER STATION

Development of a Mini Weather Station Using Electronic Sensors, Object-Oriented Programming (OOP) and Recyclable Materials

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ABSTRACT

This project presents the development of a programmed mini weather station capable of monitoring environmental variables through electronic sensors and displaying information in real time.

The system integrates an ESP32 microcontroller, Object-Oriented Programming (OOP) principles, modular programming techniques, and recyclable materials to create a low-cost educational solution for environmental monitoring.

The prototype measures temperature, relative humidity, soil moisture, light intensity, and air quality. The collected data are processed and displayed on an LCD screen, demonstrating the practical application of embedded systems and modern software development methodologies.

OBJECTIVE

General Objective

Design and implement a programmed mini weather station capable of monitoring environmental variables through electronic sensors and Object-Oriented Programming techniques.

Specific Objectives

- Measure environmental variables using electronic sensors.
- Apply Object-Oriented Programming concepts in software development.
- Integrate hardware and software into a functional embedded system.
- Promote the use of recyclable materials in technological projects.

THEORETICAL FOUNDATION

Weather Station

An automated weather station is a system designed to measure, record, and process environmental variables using electronic sensors and embedded systems.

Object-Oriented Programming (OOP)

OOP is a software development paradigm based on classes and objects. It's supported by Encapsulation, Inheritance, Polymorphism, Abstraction.

Modular Programming

It divides software into independent modules using header files (.h) and implementation files (.cpp), making code easier to maintain, test and reuse.

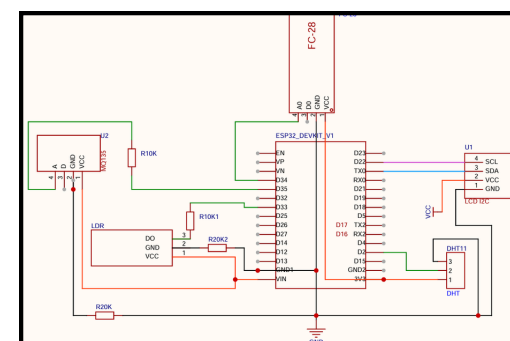
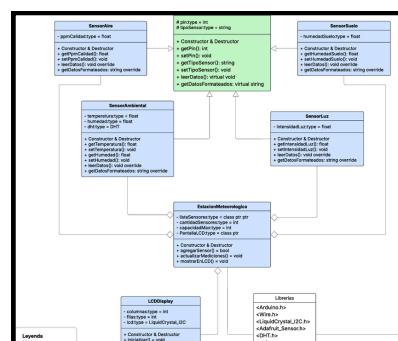
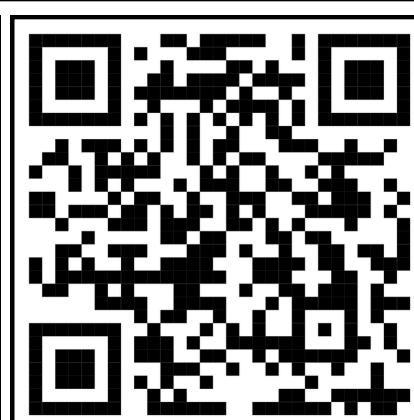
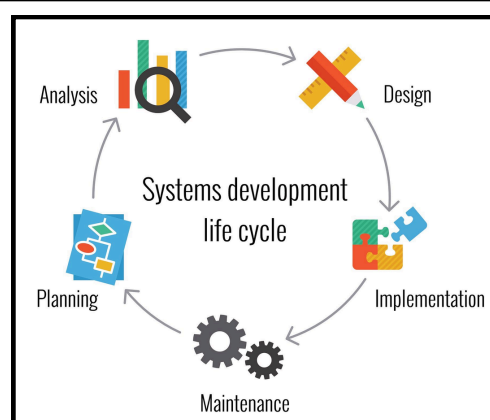
ESP32 Microcontroller

The ESP32 is a powerful microcontroller with integrated Wi-Fi and Bluetooth capabilities. It is responsible for acquiring, processing, and managing environmental data collected by the sensors.

METHODOLOGY

It was designed using an OOP approach and hardware-software integration, following:

1. **Analysis and Selection:** Identification of critical variables (temperature/humidity, soil, light, and air quality) and selection of components (ESP32, DHT11, HL-69, LDR, MQ-135).
2. **Architectural Design:** Creation and interrelation of derived classes (Sensors), a display screen (LCD I2C) and master class who manage all the programmed station functions using inheritance and polymorphism.
3. **Software Implementation:** Development in VSCode with PlatformIO/Arduino IDE framework, applying a modular architecture and encapsulation and dynamic memory.
4. **Hardware Integration:** Connection of sensors to the ESP32 microcontroller following electrical standards and configuration of the I2C bus for real-time visualization.
5. **Testing and Validation:** Unit validation of sensors, full system integration testing and stability tuning under various environmental conditions.



RESULTS

- Temperature and relative humidity measurements close to analog ambient values
- Real-time measurements and monitoring of soil moisture levels.
- Successful detection of extreme changes in ambient light levels.
- Successful measurement of the number of particles per million (PPM) present in the air.
- Displayed real-time environmental information at regular time intervals.

CONCLUSIONS

- The developed mini weather station successfully integrates multiple environmental sensors into a single monitoring platform.
- Object-Oriented Programming enabled the development of organized, modular, and scalable software.
- The use of the ESP32 and recyclable materials allowed the construction of a functional, low-cost, and sustainable prototype.
- The project demonstrates the practical application of embedded systems and software engineering concepts in real-time environmental monitoring.

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